Internship Offer: Spectral Line Analysis of OB Stars using SDSS Data

Host Lab: Space and Astrophysical Research Lab (SARL), IST
Level: BS Student (2nd year)
Team size: Two students, working in parallel on a shared pipeline
Duration: 6–8 weeks (Week 1 may extend to ~1.5–2 weeks for students new to research)
Prerequisites: See below

Overview

This internship offers a hands-on introduction to observational stellar astrophysics through the analysis of optical spectra from a sample of 100 OB-type stars drawn from the Sloan Digital Sky Survey (SDSS) Data Release 17 (DR17). Two students will each work on their own 50-star subset (randomly split to avoid spatial or brightness bias), using a shared pipeline design, and will build a systematic spectral line analysis workflow — measuring equivalent widths (EWs) and line ratios — to extract meaningful astrophysical results, including the identification of Be star candidates.

The work bridges observational data, spectral fitting, and basic Python-based scientific computing. It's designed as a first research experience: the pipeline itself is scoped to scale well beyond this project, and the supervisor will apply the validated pipeline to the full 500+-star archive after the internship concludes. By the end of the internship, each student will have contributed to a reproducible, well-documented analysis pipeline and a set of results suitable for inclusion in a research report, with the two 50-star catalogs merged as a joint final deliverable.

The 100-star combined sample is deliberately small for three reasons:

1. It keeps the project manageable within 6–8 weeks for students new to research
2. It allows both students to contribute to a complete, publishable-quality analysis without being overwhelmed
3. The validated pipeline is scoped to scale cleanly to the full 500-star archive, which the supervisor will run after the internship — meaning the students' work directly enables a larger project

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Scientific Background

OB stars are hot, massive stars at the high-temperature end of the main sequence. Their optical spectra are characterized by:

- **Hydrogen Balmer series** (Hα, Hβ, Hγ, ...) in absorption, with EW varying systematically across spectral subtypes
- **Helium lines**: He II (λ4686 Å) prominent in O stars, He I (λ5876, λ4471 Å) peaking through B subtypes — a direct spectral type diagnostic

- **Metal lines** (Si, C, Mg) at specific subtypes

A particularly interesting subclass are **classical Be stars** — rapidly rotating B stars that develop a circumstellar decretion disk, producing Hα *emission* rather than absorption. EW(Hα) flips sign in these objects, with the classical literature threshold at EW(Hα) < −5 Å. Even at the scale of this internship's 100-star combined sample, a handful of Be star candidates are plausible, making their identification a concrete, well-scoped mini-result — and one that feeds directly into the larger 500+-star analysis the pipeline is designed for.

What the Student Will Do

Phase 1 — Setup & Familiarization (Week 1, extending to ~1.5–2 weeks as needed)

- Set up the working environment (Python, `astropy`, `specutils`, `pymultifit`, `matplotlib`)
- Understand the SDSS spectral data format (FITS files, wavelength calibration, flux units) — a local archive of pre-selected DR17 spectra is provided
- Work through the pipeline spec (continuum windows, line profile choices, EW formula, output table format) together with the co-intern and the supervisor — this is designed jointly in week 1 so both students start from the same footing
- Use the golden set of 10–20 manually fitted stars as a teaching tool during this phase, not just as a later validation check. The golden set is provided with both the raw spectra and the expected EW values, so you have a clear reference to check your pipeline against as you build it.
- Review a set of pre-fitted example spectra provided by the supervisor to understand the fitting approach already established
- *(Optional)* Re-download spectra with a stricter SNR cut (e.g., SNR > 10 at 5500 Å) under supervision
- No prior research experience is assumed — this phase is intentionally unhurried, and extending it is expected and fine

Phase 2 — Continuum Normalization & Line Fitting (Weeks 2–3)

- Implement a robust continuum normalization routine across your 50-star subset (polynomial or spline fitting to pre-defined line-free windows; see Survival Guide)
- Use `pymultifit` to fit target absorption/emission lines:
  - Hα (6563 Å), Hβ (4861 Å), Hγ (4341 Å)
  - He I (5876 Å, 4471 Å)
  - He II (4686 Å) — presence/absence used as O vs. B discriminator
- Handle edge cases: blended lines, low SNR spectra, bad pixels/artifacts (see Survival Guide)
- Store fit results (amplitude, center, width, uncertainties) in a structured output table

Phase 3 — Equivalent Width Computation (Week 4)

- Implement EW calculation from fitted line parameters:
  - Lorentzian: `EW = |amplitude| × π × scale / continuum`
  - Gaussian: `EW = |amplitude| × sqrt(2π) × sigma / continuum`
- Propagate uncertainties via first-order partial derivatives:

$$\sigma_{EW}^2 = \left(\frac{\partial EW}{\partial A}\right)^2 \sigma_A^2 + \left(\frac{\partial EW}{\partial \sigma}\right)^2 \sigma_\sigma^2 + \left(\frac{\partial EW}{\partial C}\right)^2 \sigma_C^2$$

Covariance terms may be ignored at first pass.

- Use correct sign convention: **negative EW = emission**
- Build a clean per-star results table (CSV/FITS) with EWs and uncertainties for all lines

Phase 4 — Sample Analysis & Be Star Identification (Weeks 5–6)

- Produce EW distributions for each line across your 50-star subset
- Validate against known spectral-type trends (Balmer EW vs. subtype, He II presence in O stars) using the golden set introduced back in Week 1
- Flag **Be star candidates** using a two-gate criterion:
  - EW(Hα) < −5 Å (classical literature threshold; Jaschek & Jaschek convention)
  - |EW(Hα)| / σ_EW > 3 (statistical significance gate)
  - Both conditions must be satisfied; visually inspect all flagged candidates
- Compare He II detection statistics as an O vs. B subtype discriminator
- Generate population statistics: median EWs, scatter, and subtype breakdown using SDSS photometric colors

Merging the catalogs (end of Week 6):

- Each student will have a results table for their 50 stars, using the same column format and flag conventions agreed on in Week 1
- Merge these two tables into a single combined 100-star catalog
- Flag each row with the student who processed it (`analyst = "Student A"` or `"Student B"`) — this makes it easy to trace any systematic differences during validation
- Run the validation plots (EW vs. spectral type, He II detection, etc.) on the combined catalog to produce the final plots for the report

Phase 5 — Documentation & Report (Weeks 7–8)

- Write clean, commented, and reproducible Python code with a clear README
- Produce a short technical report (~10–15 pages) covering: methodology, pipeline design, results, figures, and discussion

- Prepare a brief presentation for the lab group (~15 minutes); the merged 100-star catalog can be presented jointly if the schedule allows

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Expected Deliverables

Deliverable	Description
Analysis pipeline	Well-structured, documented Python codebase (Git repository), shared between both students
Results table	Per-star EW measurements with uncertainties for all target lines (50 stars per student)
Merged catalog	Combined 100-star catalog from both students' subsets — joint deliverable
Be star candidate list	Flagged sample with EW(Hα), uncertainty, SNR, and visual inspection notes
Validation plots	EW vs. spectral type, He II detection trends, EW distributions, comparison to golden set
Technical report	Methodology, results, and discussion (~10–15 pages), individual or co-authored
Lab presentation	15-minute talk summarizing the work

What the Student Will Learn

Astrophysics:

- Physics of hot star atmospheres and spectral classification
- What equivalent width measures physically and why it matters
- The Be star phenomenon and circumstellar disk physics (introductory level)
- How to critically interpret spectral features: detections, upper limits, and SNR thresholds

Data & Methods:

- Working with real astronomical FITS spectral data (SDSS DR17 format)
- Continuum normalization techniques and their pitfalls
- Spectral line fitting with Python (`pymultifit`, `astropy`, `specutils`)
- Uncertainty propagation from fit parameters to derived quantities
- Building a pipeline that scales from one spectrum to hundreds

Software & Scientific Practice:

- Writing clean, reproducible scientific Python code
- Structured data output (tables, catalogs)
- Scientific visualization (`matplotlib`)
- Version control basics (Git)
- Technical scientific writing

Prerequisites

Required:

- Comfort with Python (NumPy, matplotlib, basic file I/O)
- Completed or currently enrolled in at least one Astrophysics or Physics course
- Willingness to read and understand scientific literature at an introductory level

Self-assessment exercise (recommended before applying):
Using `astropy.io.fits`, read a FITS spectrum, plot flux vs. wavelength, and fit a Gaussian to a single isolated line using `scipy.optimize.curve_fit`. This is meant as a gauge, not a gate — if parts of it feel unfamiliar, that's expected for a first research project, and Week 1 is built to bring both students up to speed together.

Helpful but not required:

- Prior exposure to FITS files or `astropy`
- Familiarity with spectroscopy concepts
- Experience with curve fitting in Python

Supervision & Support

Both students will be directly supervised at SARL, with the pipeline design worked out jointly in Week 1 so neither student is figuring things out in isolation. Weekly check-ins will cover progress, debugging, and results discussion, and can be done together where useful (e.g., the final catalog merge). A golden set of 10–20 manually fitted spectra will be provided at the start and used both as a teaching aid and for later pipeline validation. As this may be a first research experience for both students, the pace in Week 1 is flexible, and questions or blockers are expected and welcome rather than something to work around alone.

Collaboration & Code Review

The two students will work on their own 50-star subsets, but the underlying codebase is shared. To keep things organized:

- Use a shared Git repository (GitHub or GitLab) with separate branches or a clear folder structure for each student's analysis scripts

- Review each other's code at least once before merging changes — a lightweight code review catches bugs and builds confidence in the pipeline
- Agree on column names and flag conventions in Week 1 (e.g., `flag = "low_SNR"`, `flag = "fit_failed"`) so the final 100-star merge is seamless
- Hold a short sync with the co-intern at the end of Week 2 to compare continuum normalization approaches and ensure consistency before scaling to the full 50-star set — it's easier to fix disagreements early, and again at the end of Week 4 before merging catalogs, to confirm flag conventions and column formats are still aligned

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Notes

- All data is already available (local FITS archive from SDSS DR17) — no data acquisition overhead
- The pipeline is built on top of `pymultifit`, an in-house open-source Python library actively maintained at SARL; students gain direct exposure to a real PyPI package
- The 100-star combined sample is a deliberately manageable first step — the validated pipeline will later be applied by the supervisor to the full 500+-star archive
- Strong performance may lead to co-authorship consideration on a future publication (e.g., a validated EW catalog used in a paper) or a follow-up MS internship offer

Flexibility note: If the 8-week timeline feels tight — especially if Week 1 extends to 2 weeks — the priority will be completing a clean, validated 50-star subset rather than rushing through all phases. The supervisor can help adjust the scope (e.g., dropping He II or a subset of metal lines) to ensure the core deliverables are high quality. The goal is a solid research experience, not perfection.

Document prepared by SARL, IST